

CMT Problem Set 2 Homework #2

Problem 2.1 (Normal modes of a square lattice). Consider a square lattice of atoms of mass M , in which the nearest neighbors are coupled to each other by springs with constants K . Derive the equation of motion for the mode in which atomic displacements are normal to the crystal plane and find the dispersion of this mode. Analyze the behavior of this mode in the continuum limit.

Solution. Assume the lattice constant a , then the lattice sites become

$$R_{n,m} = (na, ma), \quad n, m \in \mathbb{Z},$$

We consider every atom has a single degree of freedom: the out-of-plane scalar displacement $U_{n,m}(t)$. So, the spring connecting site (n,m) and a neighbour (n',m') has extension $\Delta = U_{n',m'} - U_{n,m}$, which brings the force to atom (n,m)

$$F_{n,m} = -K(U_{n,m} - U_{n',m'}),$$

due to Hooke's law. Since every atom has four nearest neighbours: $(n \pm 1, m)$, $(n, m \pm 1)$, which bring four forces to the "Central Atom".

Summing these forces

$$F_{n,m}^{\text{tot}} = -\sum_{\langle n,m; n',m' \rangle} K(U_{n,m} - U_{n',m'}) = -K(4U_{n,m} - U_{n+1,m} - U_{n-1,m} - U_{n,m+1} - U_{n,m-1})$$

According to Newton's 2nd law, we obtain the EOM

$$M\ddot{U}_{n,m} = -K(4U_{n,m} - U_{n+1,m} - U_{n-1,m} - U_{n,m+1} - U_{n,m-1}).$$

Assume the EOM has a plane-wave solution

$$U_{n,m}(t) = Ae^{i(k_x na + k_y ma - \omega t)},$$

then substitute it into the EOM, the LHS and RHS are

$$\text{LHS} = M(-\omega^2)U_{n,m}; \quad \text{RHS} = -2K[2 - \cos(k_x a) - \cos(k_y a)]U_{n,m}.$$

Finally we have the dispersion of this model

$$\omega(\vec{k}) = 2\sqrt{\frac{K}{M}} \sqrt{\sin^2\left(\frac{k_x a}{2}\right) + \sin^2\left(\frac{k_y a}{2}\right)}$$

In the continuum limit, i.e., $|\vec{k}|a \ll 1$, then the dispersion of this model becomes linear to $|\vec{k}|$

$$\omega(\vec{k}) \cong 2\sqrt{\frac{K}{M}} \cdot \sqrt{\left(\frac{k_x a}{2}\right)^2 + \left(\frac{k_y a}{2}\right)^2} = c|\vec{k}|$$

where $c = a\sqrt{K/M}$.

Problem 2.2 (Localized phonons on an impurity). Consider a linear chain of identical atoms of mass m each equidistant by a .

- Limiting the problem to interactions between nearest neighbors characterized by the force constant β , give the dispersion relation of longitudinal phonons and derive the expression of their group velocity. Express the results as a function of " a " and of the velocity of sound, v_s . What is maximum frequency of atomic vibration, ω_M , when $v_s = 5000 \text{ m/s}$?
- An atom in the middle of the chain is replaced by the atom of an impurity with a mass m' (where $m' < m$). Write the displacement equations, u_0 , for this impurity atom and the displacement " u_n " for an atom located at a distance " na ". Assume that the force constant β between nearest neighbors is the same regardless of the atom considered.
- The impurity atom being lighter than the others has an oscillation frequency ω'_m larger than ω_M and it drags the other neighbor atoms by its motion. We try to describe its effect with the expression of a damped wave of form: $u_n = u_0 e^{-\alpha|x|} e^{i(\omega't - kx)}$. What is the value of the frequency ω'_M , which is compatible with the equations of motion when the wave vector is near the boundary of the first Brillouin Zone? Verify that when $m = m'$, the solution matches that obtained in (a).
- After having numerically evaluated ω'_M and α when $m' = m/2$, sketch the instantaneous position of atoms as a function of x .

Solution.

(a) Assume the nearest neighbour spring constant β . The EOM for the n -th atom is

$$m\ddot{u}_n = -\beta(u_n - u_{n+1}) - \beta(u_n - u_{n-1}) = \beta(u_{n+1} + u_{n-1} - 2u_n)$$

We assume the EOM has a plane-wave solution $u_n(t) = A e^{i(kna - \omega t)}$, then substitute it into the EOM

$$-m\omega^2 u_n = -\beta(2 - e^{ika} - e^{-ika}) u_n = -\beta(2 - 2\cos(ka)) u_n = -4\beta \sin^2\left(\frac{ka}{2}\right) u_n$$

we have the dispersion

$$\omega = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

The group velocity and the velocity of sound are

$$v_g(k) = \frac{d\omega}{dk} = a\sqrt{\frac{\beta}{m}} \cos\left(\frac{ka}{2}\right), \text{ and } v_s(a) = \lim_{k \rightarrow 0} v_g(k) = a\sqrt{\frac{\beta}{m}}$$

then the group velocity, as well as the frequency can be expressed as

$$v_g(k) = v_s \cos\left(\frac{ka}{2}\right), \text{ and } \omega = \frac{2v_s}{a} \left| \sin\left(\frac{ka}{2}\right) \right|$$

when $ka = \pi$, the frequency arrives at a maximum value

$$\omega_M = \frac{2v_s}{a} \frac{v_s = 5000 \text{ m/s}}{10000} \frac{10000}{\text{cm}} \text{ s}^{-1}$$

(b) The impurity is located at site $n=0$, with mass m' ; while its neighbours $n=\pm 1$ with mass m . The EOM for the impurity and other atoms are

$$m' \ddot{u}_0 = -\beta(u_1 - u_0) + \beta(u_{-1} - u_0) = \beta(u_1 + u_{-1} - 2u_0)$$

$$m \ddot{u}_n = \beta(u_{n+1} + u_{n-1} - 2u_n)$$

(c) In (a), near the boundary of the 1st B.Z., i.e., $k=\pi/a$, the frequency is $\omega_I = 2\sqrt{\beta/m}$. Write the given wave form in terms of site No. n

$$u_n = A e^{-\alpha|n|a} e^{i(\omega t - kna)}$$

Substitute it into the two EOMs in (b), and let $n=1$

$$-\omega^2 m' u_0 = \beta(e^{-\alpha a - ika} + e^{-\alpha a + ika} - 2)u_0 \xrightarrow{k=\pi/a} 2\beta(e^{-\alpha a} + 1)u_0$$

$$-\omega^2 m u_1 = \beta(u_2 + u_0 - 2u_1) = \beta(e^{-\alpha a - ika} + e^{\alpha a + ika} - 2)u_1 \xrightarrow{k=\pi/a} -2\beta(\cosh(\alpha a) + 1)u_1$$

Derive the two expressions, we have

$$\frac{m'}{m} = \frac{e^{-\alpha a} + 1}{\cosh(\alpha a) + 1} = \frac{2}{e^{\alpha a} + 1}, \text{ and } e^{-\alpha a} + 1 = \frac{2m}{2m - m'}$$

Substitute it into one of the EOMs, we obtain the dispersion

$$\omega_m'^2 = \omega^2 = \frac{4\beta m}{m'(2m - m')}$$

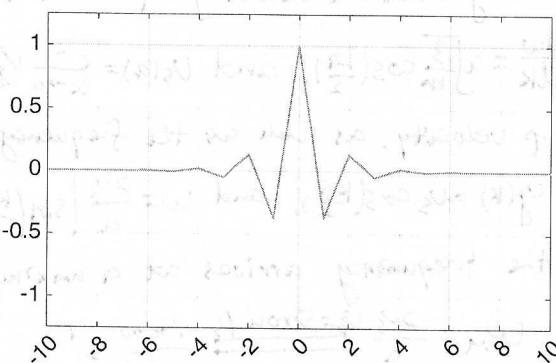
where $\omega_m'^2 = \omega^2$ is due to the boundary of 1st B.Z. ($k=\pi/a$). When $m'=m$, the frequency becomes

$$\omega_m'^2 = 4\beta/m, \text{ and } \omega_m' = 2\sqrt{\beta/m} = \omega_I$$

which matches that obtained in (a).

(d) When $m'=m/2$, $\omega_m'^2 = 16\beta/3m$. To simplify, we choose the time $t=0$. Near the 1st B.Z., the displacement of atoms is

$$u_n = (-1)^n A e^{-\alpha|n|a}$$



* The plotted figure of atoms only show the relative position for reference.

Plotted by MATLAB

Problem 2.3 (Density of states and specific heat of a monoatomic 1D lattice). Consider a row of N identical atoms of mass m and equidistant by a .

- Limiting the problem to interactions between nearest neighbors with a force constant β , find the expression for the dispersion relation of acoustic longitudinal vibrations propagating along this row. What is the maximum frequency ω_M of atomic vibrations, written in terms of v_s and a ? Compare the value obtained with that of ω_D deduced from the Debye model.
- Find the corresponding expression for the density of states $g(\omega)$.
- Write the expressions for internal energy U and for the heat capacity C_v . Find the limits as these functions approach high temperatures ($\hbar\omega_M \ll k_B T$).
- What is the evolution of C_v at low temperature? Compare the result obtained here with that deduced from the Debye model.

Solution.

(a) The same to problem 2-2(a): the dispersion

$$\omega(k) = 2\sqrt{\frac{\beta}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|, \quad v_s = \lim_{k \rightarrow 0} \frac{d\omega}{dk} = a\sqrt{\frac{\beta}{m}}, \quad \text{and } \omega_M \xrightarrow{k=\pi/a} 2\sqrt{\frac{\beta}{m}} = \frac{2v_s}{a}$$

and the frequency for 1D chain in Debye model is $\omega_D = v_s k = \frac{\pi v_s}{a}$. Since $\frac{\omega_M}{\omega_D} = \frac{2}{\pi} < 1$, the maximum frequency in the real dispersion is smaller than in the Debye model.

(b) In 1D chain, the number of states within dk is $2 \cdot \frac{L}{2\pi} \cdot dk$. So, the density of states is

$$g(\omega) = \frac{L}{\pi} \frac{dk}{d\omega} = \frac{2L}{\pi a \omega_M} \frac{1}{\sqrt{1-(\omega/\omega_M)^2}} \xrightarrow{L=N a} \frac{2N}{\pi \omega_M} \frac{1}{\sqrt{1-(\omega/\omega_M)^2}}.$$

(c) The phonon thermal occupation

$$n(\omega) = \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1}$$

Then the internal energy is

$$U(T) = \int_0^{\omega_M} \hbar\omega n(\omega) g(\omega) d\omega = \int_0^{\omega_M} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} g(\omega) d\omega$$

and the heat capacity

$$C_v(T) = \frac{dU}{dT} = \int_0^{\omega_M} \frac{\partial n(\omega)}{\partial T} g(\omega) d\omega$$

At high temperature limit, i.e., $\hbar\omega_M \ll k_B T$. The two expressions above becomes

$$U(T) = \int_0^{\omega_M} k_B T g(\omega) d\omega = N k_B T, \quad C_v(T) = N k_B.$$

where applied the total number of vibration modes $\int_0^{\omega_M} g(\omega) d\omega = N$.

(d) At low temperature, only small k matters, i.e., the main contributions come from small w . Then, $g(w) = \frac{L}{\pi v_s}$. The internal energy becomes,

$$U = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar w}{e^{\frac{\hbar w}{k_B T}} - 1} dw = \frac{\pi L (k_B T)^2}{6 \hbar v_s}$$

and the capacity

$$C_V = \frac{dU}{dT} = \frac{\pi L k_B^2 T}{3 \hbar v_s} \propto T$$

the same conclusion as the Dybye model.

(a) The same to problem 2 with the observation

$$\frac{dU}{dT} = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar w}{e^{\frac{\hbar w}{k_B T}} - 1} dw = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar w}{e^{\frac{\hbar w}{k_B T}} - 1} dw = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar w}{e^{\frac{\hbar w}{k_B T}} - 1} dw$$

and the frequency for 1D chain is $\omega = \frac{v_s}{L} k$, so $\frac{dU}{dT} = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} d\omega = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} d\omega$

the minimum frequency in the real dispersion is smaller than in the Dybye model.

(b) In 1D chain, the number of states within dk is $\frac{L}{2\pi} dk$. So, the density of states

$$g(\omega) = \frac{1}{2\pi} \frac{dN}{d\omega} = \frac{1}{2\pi} \frac{dN}{dk} \frac{dk}{d\omega} = \frac{1}{2\pi} \frac{L}{v_s} \frac{1}{\omega}$$

(c) The phonon thermal occupation

$$n(\omega) = \left[\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1 \right]^{-1}$$

Then the internal energy is

$$U(T) = \int_0^{\infty} \hbar \omega g(\omega) n(\omega) d\omega = \int_0^{\infty} \hbar \omega g(\omega) \left[\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1 \right]^{-1} d\omega$$

only the heat capacity

$$C_V(T) = \frac{dU}{dT} = \frac{L}{\pi v_s} \int_0^{\infty} \frac{\hbar \omega}{T^2} \left[\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1 \right]^{-2} d\omega$$

At high temperature limit, $k_B T \gg \hbar \omega$, so $\exp\left(\frac{\hbar \omega}{k_B T}\right) \approx 1 + \frac{\hbar \omega}{k_B T}$. The two expressions above become

$$U(T) = \int_0^{\infty} \hbar \omega g(\omega) n(\omega) d\omega = \int_0^{\infty} \hbar \omega g(\omega) \left[\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1 \right]^{-1} d\omega$$

where applied the total number of vibration modes $2N$, $\int_0^{\infty} g(\omega) d\omega = 2N$